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Evaluating the detection limit of organic matter using point and imaging spectroscopy

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ABSTRACT

The commonly used optical technique for estimating organic matter in soils is based on a chemometric approach. Many studies have examined the correlation between spectral response and organic matter content, but none have investigated uncertainties based on unequal spectral responses in various soil types or spectral detection limit. The aim of this study was to systematically examine the spectral responses of five different soils with increasing amounts of organic matter to evaluate detection limits in point (ASD) and image (HySpex) domains. In addition, we evaluated the contribution of clay content and soils' initial organic matter on the assessed detection limit for each sensor. Large spectral variations were found between soils with the same organic matter content and between the two sensors when calculating the detection limit. Thus, applying a generic prediction model of organic matter using all soil types results in a rough estimation due to the spectrum affiliation with a certain class and not due to correct organic matter analysis. The importance of this study lies in showing that the use of spectroscopy for spectral-based organic matter detection should be practiced with caution, by highlighting the variant and inconsistent effects of organic matter in soils, and discussing the problematic assessment of organic matter when using a chemometric approach.

1. Introduction

Soil organic matter (SOM) includes all living and dead organisms within the soil matrix. It is responsible for supplying nutrients to vegetation and generating aggregates by binding soil particles together (Ashman and Puri, 2013), and it is one of the five state factors in soil formation (Buol et al., 2011). In addition, it influences other soil properties, such as compaction, water retention and soil-structure stability (Ben-Dor et al., 2009; Stevens et al., 2008), darkens soil color (Chomchan et al., 1979), and is one of the main indicators for soil-quality assessment (Parr et al., 1992; Reeves, 1997; Visser and Parkinson, 1992). SOM is one of the key criteria for the histic, folistic, mollic and umbric epipedons in the United States Department of Agriculture (USDA) classification (Bockheim, 2014a, 2014b; Soil Survey Staff, 2011) and as a diagnostic horizon in the World Reference Base (WRB) system (IUSS Working Group, 2015); it is also used to differentiate organic soils (e.g. histosols) from mineral soils (e.g. vertisols) (Buol et al., 2011), as it can range from < 0.1% in desert soils to close to 100% in organic soils (Schnitzer and Khan, 1975). Most of the soils in Israel contain low organic matter (OM) content, although some soils, such as Terra Rosa and Mediterranean brown forest soils (see Table 1 for the equivalent WRB name), may contain higher levels of OM (Ravikovitch, 1992). Several traditional laboratory analyses exist for SOM assessment: oxidation of OM with hydrogen peroxide (Gallardo et al., 1987), the Walkley–Black titration method in which OM is oxidized with potassium dichromate in sulfuric acid (Nelson and Sommers, 1982; Walkley and Black, 1934), and the loss-on-ignition (LOI) method (Ben-Dor and Banin, 1989). However, these methods are either expensive or time-consuming.

In the past few decades, there have been many studies devoted to investigating the correlation between spectral data in the visible, near-infrared and shortwave-infrared (VNIR–SWIR) regions (400–2500 nm) and OM content or its correlated attribute organic carbon using the spectroscopy and imaging spectroscopy domains (Ben-Dor and Banin, 1995; Ben-Dor et al., 1997; Ben Dor et al., 1999; Daniel et al., 2003; Hummel et al., 2001; Jin et al., 2016; Sudduth and Hummel, 1993; Krishnan et al., 1980; Lamsal, 2009; Liu et al., 2008; Peón et al., 2017; Shi et al., 2015; Tian et al., 2013; Volkan Bilgili et al., 2010; Wang et al., 2016). Although many studies use the entire spectral range, some have found specific spectral ranges for chemometric modeling, mostly

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Table 1
The soil types of the Israeli soil classification system with the equivalent World Reference Base (WRB) system, their texture and OM properties

Code	Soil series Israel	WRB	Clay (%)	Silt (%)	Sand (%)	OM_S (% wt)
H	Hamra (brown-red sandy soils)	Rhodic Luvisols	0	0	100	0.61
L	Loess	Luvic Calcisols	9.8	43.6	46.6	4.13
R	Rendzina	Calcaric Leptosols	14.5	53.5	32.0	5.23
T	Terra Rosa	Epileptic Luvisols	17.1	59.9	23.0	9.01
S	Sand (sand dunes)	Arenosols	0	0	100	0.27
	Compost	–	–	–	–	35.8

from 400 to 1000 nm (Cozzolino and Morón, 2006; Henderson et al., 1992; Krishnan et al., 1980; Tian et al., 2013; Vaudour et al., 2016). Numerous studies have examined the linear fitting between OM and spectra, but none has investigated how different contents of OM affect the prediction model in different soil types, nor have any taken into consideration the detection limit (DL) of the spectral-based approach to determine OM content with reliable results. The DL, or limit of detection (LoD), is defined as the lowest detectable amount of analyte in a sample in a given method (Broughton and Ermer, 2005). It is expressed as the analyte concentration corresponding to the sample < blank > + 3 standard deviations (σ) (Shrivastava and Gupta, 2011).

Since different soils exhibit different spectral responses due to their chemical and physical compositions, the effect of the same OM content is assumed to be variant and inconsistent. As a result, a chemometric approach to evaluating SOM cannot be successful using generic statistical approaches, and the DL of OM for every soil type must be accurate.

In this study, five different soil types from different locations in Israel were collected. Each soil was air-dried, ground and sieved to 2 mm and then was divided into 18 sub-samples which contain varying percentages of OM (an industrial compost—used as an indicator of OM content). The compost was added in small intervals of 0.2% until it reached a total of 2%, which is sufficient for identifying changes in each individual spectrum. In order to achieve commonality and compare between soils with low OM (Sand and Hamra) and soils with high OM (Terra Rosa) in a joint model, six additional sub-samples with higher percentages of compost were added (2.5, 3, 5, 11.1, 20, 33.3 and 42.9%). The main objective of this study was to systematically examine the effect of varying amounts of a known OM contents to evaluate the soil spectral detection limit (SSDL), using both point and imaging spectroscopy under controlled laboratory conditions.

2. Materials and methods

2.1. Soil samples and laboratory measurements

Five different soil types were collected in Israel from the surface level. In addition, an industrial compost material was used as an OM demonstrator source which is added to the initial OM content. All samples were air-dried, ground and sieved to 2 mm. The soils' texture was measured with a laser diffraction particle-size analyzer (Mastersizer 2000, www.marlven.com) and the natural (initial) SOM content (OM_S) was measured by LOI (400 °C for 8 h) method (Ben-Dor and Banin, 1989; Davies, 1974). Table 1 provides the clay, silt and sand distribution and OM_S measured by the LOI method for each soil type.

2.2. Sample preparation

Each soil was divided into 18 sub-samples of 30 g each with different percentages of added compost. Distilled water was added to the sample until saturation, to enable better soil–OM mixture, stirred and then air-dried and mashed to 2 mm. Fig. 1 presents the percentage of the added compost (OM_C), the initial organic matter in the soils (OM_S) (where $OM_C = 0$), and the total organic matter content (OM_{SC}) originated from both the soil and the compost for each soil type. For weight measurements, we used an Electronic Semi-Microbalance R200D (www.sartorius.com) with standard deviation of less than ± 0.00002 g.

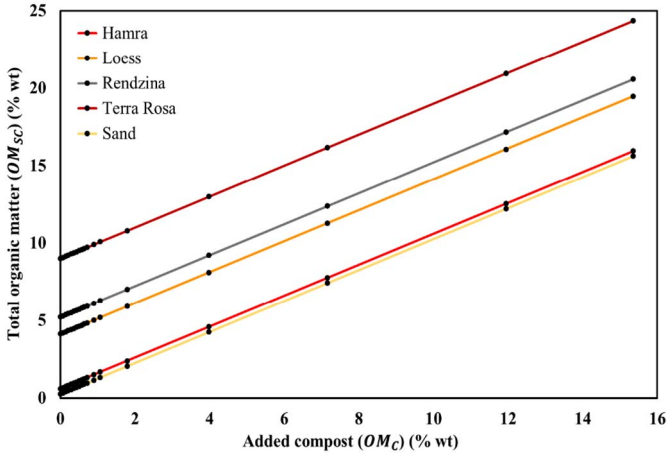


Fig. 1. The added compost (OM_C), the initial OM content in soil (OM_S) (marked as the first sample in each series) and the total OM content in the soil samples (OM_{SC}).

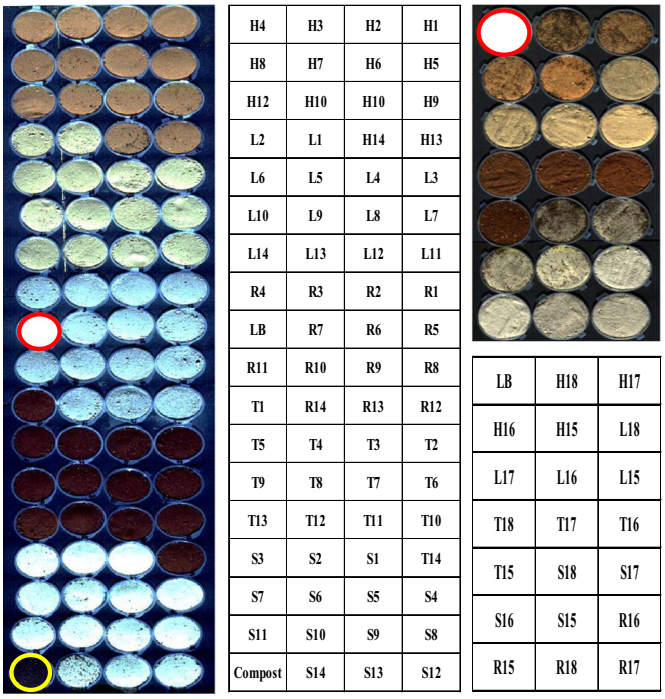
2.3. Sensor configuration and spectral data correction

The spectrum of each soil sample was measured using both point and image spectrometers using the standards and protocols proposed by Ben Dor et al. (2015). The point spectrometer was the portable Analytical Spectral Devices spectrometer ASD FieldSpec in the Remote-Sensing Laboratory at Tel Aviv University, Israel. The spectrometer has three detectors with 1 nm interpolated spectral resolution that supply 2151 bands in the VNIR (350–1000 nm), SWIR1 (1000–1800 nm) and SWIR2 (1800–2500 nm) regions.

The hyperspectral images were acquired at the Remote-Sensing Laboratory of GFZ German Research Centre For Geosciences, Potsdam, Germany, using the HySpex imaging spectrometer. The sensor contains two detectors in the VNIR range and SWIR range with spectral resolution of 3.6 and 6 nm, respectively. The integration time was adjusted manually so that the maximum radiance within all samples could be covered without saturation. Subsequently, the total 436 bands (between 400 and 2500 nm) were reduced to 399 bands due to extraction of overlapping wavelengths. Additional information regarding the soil measurement and spectral correction procedures are can be found in Rogass et al. (2017).

Fig. 2 shows the HySpex images together with the spatial arrangement of the samples. The orange circles mark the internal soil standard as proposed by Ben Dor et al. (2015) and the yellow circle marks the compost sample.

Following the initial preparation, samples were measured using a contact probe equipped with a halogen bulb and Spectralon® as a white reference and corrected using the internal soil standard as discussed in Ben Dor et al. (2015). Due to data acquisition differences between the sensors (point vs. pixels), different approach for data collection needed to be taken. Using the ASD, each soil type was measured 3 times (to get the average spectrum of the sample) while first sample (compost-free sample or blank) of each soil type was measured 20 times to get the mean and standard deviation for a later DL calculation. However, the image acquisition was performed only once, so in order to obtain the 20 spectra of the blank sample, the average spectrum and the standard deviation were calculated from 20 regions of interest (ROI) of 5 pixels each which resemble 20 independent measurements. The average



H4	H3	H2	H1
H8	H7	H6	H5
H12	H10	H10	H9
L2	L1	H14	H13
L6	L5	L4	L3
L10	L9	L8	L7
L14	L13	L12	L11
R4	R3	R2	R1
LB	R7	R6	R5
R11	R10	R9	R8
T1	R14	R13	R12
T5	T4	T3	T2
T9	T8	T7	T6
T13	T12	T11	T10
S3	S2	S1	T14
S7	S6	S5	S4
S11	S10	S9	S8
Compost	S14	S13	S12

LB	H18	H17
H16	H15	L18
L17	L16	L15
T18	T17	T16
T15	S18	S17
S16	S15	R16
R15	R18	R17

Fig. 2. HySpex images of the soil samples containing various amounts of compost where each sample is approximately 6 cm in diameter.

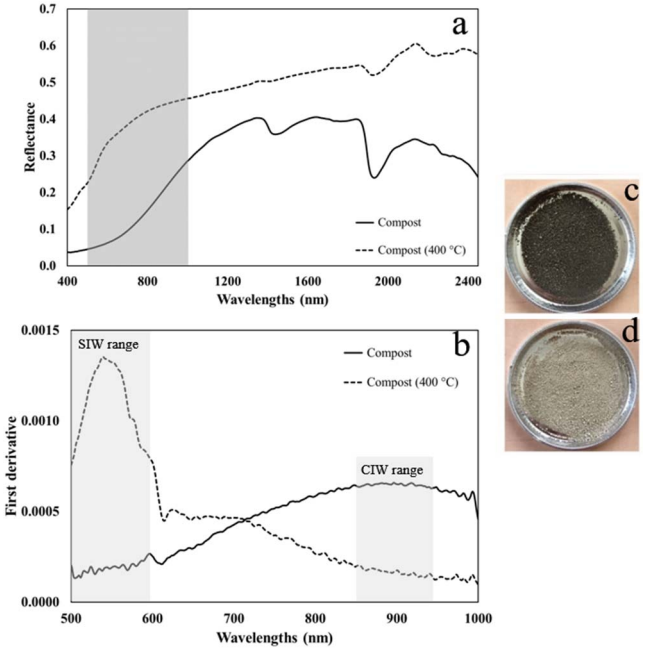


Fig. 3. Reflectance spectrum of the compost sample before and after LOI (a), the first-derivative spectra of the reflectance (between 500 and 1000 nm – the gray area in panel a) before and after LOI (b) and photos of the compost before (c) and after (d) LOI.

spectrum of the non-blank samples, on the other hand, was obtained from one ROI of the entire sample. After the spectral libraries of the two datasets were established, only the high signal-to-noise ratio bands (400–2450 nm) were chosen for the proposed spectral analysis.

2.4. Determination of the indicative wavelengths

To detect the influence of OM on the soil spectra, we had to identify the significant wavelengths that enhance the effect. Ben-Dor et al. (1997) showed that the spectral slopes in the VNIR–SWIR are basic parameters for monitoring OM, and we therefore used the Savitzky-Golay polynomic approximation (first derivative spectra) for analysis (Savitzky and Golay, 1964). The compost sample was used as a case study which was measured spectrally before and after the LOI method. Fig. 3a shows the reflectance spectra and 3b the first-derivative spectra (between 500 and 1000 nm) of the compost sample before and after oven-drying (400 °C for 8 h), and Fig. 3c and d show the compost sample before and after oven-drying, respectively. The difference between the spectral responses was due to the presence of OM, since the original compost consisted of both minerals and OM whereas after oven-drying, it consisted only of the mineral components.

The wavelengths that exhibited the most indicative spectral shift due to OM variability showed an opposite trend between the VIS and NIR regions. We could therefore conclude that the opposite spectral curve occurs due to the presence of OM and any change in OM content will be noticeable in these spectral regions. In other words, high OM content resulted in lower values in the VIS spectral range accompanied by higher values in the NIR in the first-derivative spectrum. Between 500 and 600 nm, we identified the soil indicative wavelength (SIW) range, consisting of the specific SIW for each soil type where the maximum spectral slope in a reflectance spectrum is expressed as a maximum value in the first-derivative spectrum. In contrast, the compost indicative wavelength (CIW) range, which contains the maximum value of the compost sample appeared between 850 and 950 nm (see the opposite trend in Fig. 3b).

The CIW was identical for all soil types because the same compost was used for all soil types, however, the maximum slope (highest first-derivative value) in the SIW range was located in a different wavelength. For more detailed information regarding the SIW/CIW ratio see section 3.1.

2.5. Preprocessing and spectral analysis

To assess the SSDL of OM, a new approach was developed which takes into account both the spectral variance of the selected soil types and the compost effect. After acquiring the spectral data, the following stages were applied:

i. Savitzky–Golay (S–G) smoothing, a first derivative (Mercer, 2014) and a secondary S–G smoothing were applied (order: 3; window: 11) and the 500–1000 nm range was chosen.
ii. The soil indicative wavelength (SIW) was identified for each soil type.
iii. The compost indicative wavelength (CIW) was identified.
iv. The SIW/CIW ratio was calculated for each sample.
v. The mean ($\bar{x}$) and standard deviation (σ) of the SIW/CIW ratio for the blank measurements of each soil type were calculated.
vi. The first 10 samples (< 2% of compost) of each soil series were chosen and a linear correlation curve was created between the SIW/CIW ratio and the added compost amounts.
vii. The coefficient of determination (R^2) between the SIW/CIW ratio and the added compost amounts was calculated.
viii. The eight remaining samples were added while continuously measuring the R^2.
ix. The linear equation was selected when the highest correlation was obtained using the selected number of samples.
x. The SSDL was calculated using Eq. (1) and is graphically explained in Fig. 3a:

$$SSDL_{compost}\ (\%wt) = a(\bar{x} - 3\sigma)_{< blank >} + b \quad (1)$$

where $SSDL_{compost}$ is the soil spectral detection limit of compost (which can then be translated to OM_{SC} content), a and b are the designated constants, and $\bar{x}$ and σ are the calculated mean and standard deviation of the SIW/CIW ratio of the blank sample.

In addition, we examined whether the choice of a linear model

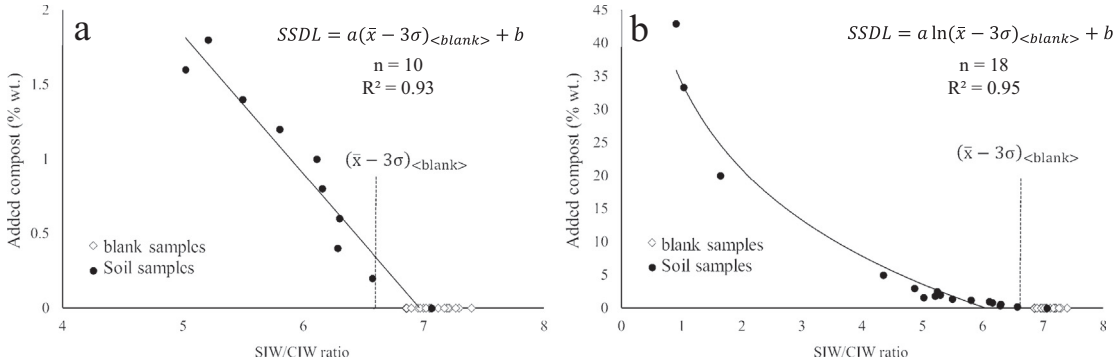


Fig. 4. Comparison between the suggested method using the highest linear correlation achieved with the first 10 samples (a) and a logarithmic correlation using all 18 samples (b) in Hamra soil. The white diamond shapes are the 20 repetitive measurements of the blank.

(≥10 samples) is preferred to a logarithmic model (18 samples) for calculation and SSDL analysis. Fig. 4a shows the linear trend achieved by the first 10 samples (0–1.8% wt of added compost), which resulted in $R^2 = 0.93$. Fig. 4b shows the logarithmic trend using 18 samples (0–42.9% wt of added compost), which resulted in $R^2 = 0.95$. Although, the last offered a higher R^2, it interacts with the x-axis before reaching the DL threshold of $(\bar{x} - 3\sigma)_{<blank>}$. As a result, the outcome yields a negative value for SSDL and therefore, we continued with the linear correlation for the SSDL calculation.

The next step is to calculate the SSDL of OM content ($SSDL_{OM}$) and the $SSDL_{OM}$anomaly, which are defined in Eqs. (2) and (3), respectively:

$$SSDL_{OM}\ (\%wt) = SSDL_{compost} * 0.358 \pm OM_S \quad (2)$$

$$SSDL_{OM}\ \ anomaly\ (\%wt) = OM_S \pm SSDL_{OM} \quad (3)$$

where $SSDL_{compost}$ is multiplied by the percent of OM in the compost sample, OM_S is the initial OM content in each soil and $SSDL_{OM}$ anomaly is the difference between the OM_S and the $SSDL_{OM}$.

3. Results and discussion

3.1. The relationship between SIW and CIW

All soils exhibited high values between 500 and 600 nm (SIW range), followed by low values between 800 and 1000 nm (CIW range); however, each soil type had its unique SIW in the SIW range. The CIW was identified using the compost sample and its value was taken from the soil spectra. Fig. 5 shows the variability in SIW positions, which are dependent on soil type and sensor, and the position of CIW, in both the ASD (Fig. 5a) and HySpex (Fig. 5b) data. A summary of the specific SIW and CIW positions is given in Table 2.

The variation between soil types was expressed not only by their different SIW positions, but also by the effect of compost amount on them. Fig. 6 presents the Hamra, Loess and Sand samples, which exhibited a rapid decrease in the SIW/CIW ratio, compared to Rendzina and Terra Rosa with only a slight decrease in ratio.

Table 2
Summary of SIW and CIW positions for soils and compost as measured with the ASD and HySpex sensors.

	Soil type	Indicative wavelength (nm)	
		ASD	HySpex
SIW	Hamra	569	560
	Loess	560	553
	Rendzina	555	546
	Terra Rosa	571	557
	Sand	542	538
CIW	Compost	897	891

3.2. Soil spectral detection limit (SSDL) assessment

The SSDL approach takes into account the ratio between the SIW and the CIW for each soil type separately, and the performances of both linear and non-linear models were investigated (Section 2.5.). Non-linearity occurred when higher amounts of compost were added to the soil samples but there was no change in the spectral response due to OM_{SC} spectral "saturation". In other words, from a certain amount of the OM_{SC} onward, the soil's spectrum appeared to be similar to the compost spectrum and adding more compost would not affect it. Therefore, a linear correlation between the SIW/CIW ratio and compost was expected until 'saturation' occurred. Thereafter, we chose the number of samples in each soil type that delivers the highest R^2 to find the linear equation for calculating its DL.

Examination of the correlation results of the ASD data (Table 3a) showed that the Hamra soil reaches its highest R^2 value when using the first 10 samples, which account for 0.0–1.8% of compost and maintain a relatively consistent R^2 (> 0.8), although additional samples were added to the calculation. The highest R^2 for Sand was reached with 11 samples (0.0–2.0% of compost) and it showed a sharp decrease from sample 14, until it reaches an R^2 of 0.4. Loess reached the highest R^2 in the 13th sample and thereafter decreased moderately, whereas both Rendzina and Terra Rosa reached the highest R^2 using 18 samples

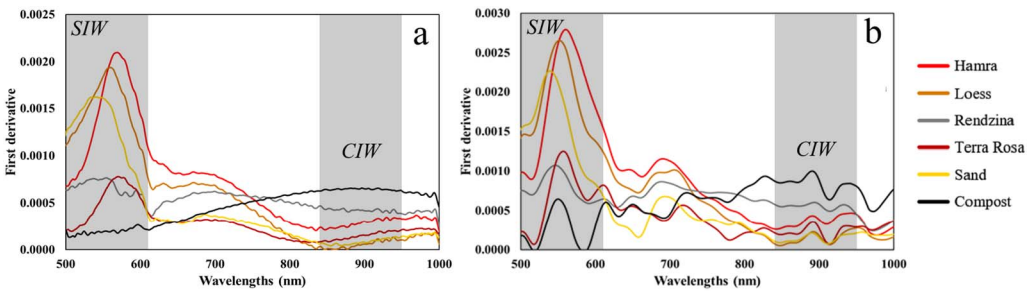


Fig. 5. First-derivative spectra between 500 and 1000 nm of the natural soils and the compost sample. The SIW and CIW ranges using the ASD (a) and HySpex (b) are marked in gray.

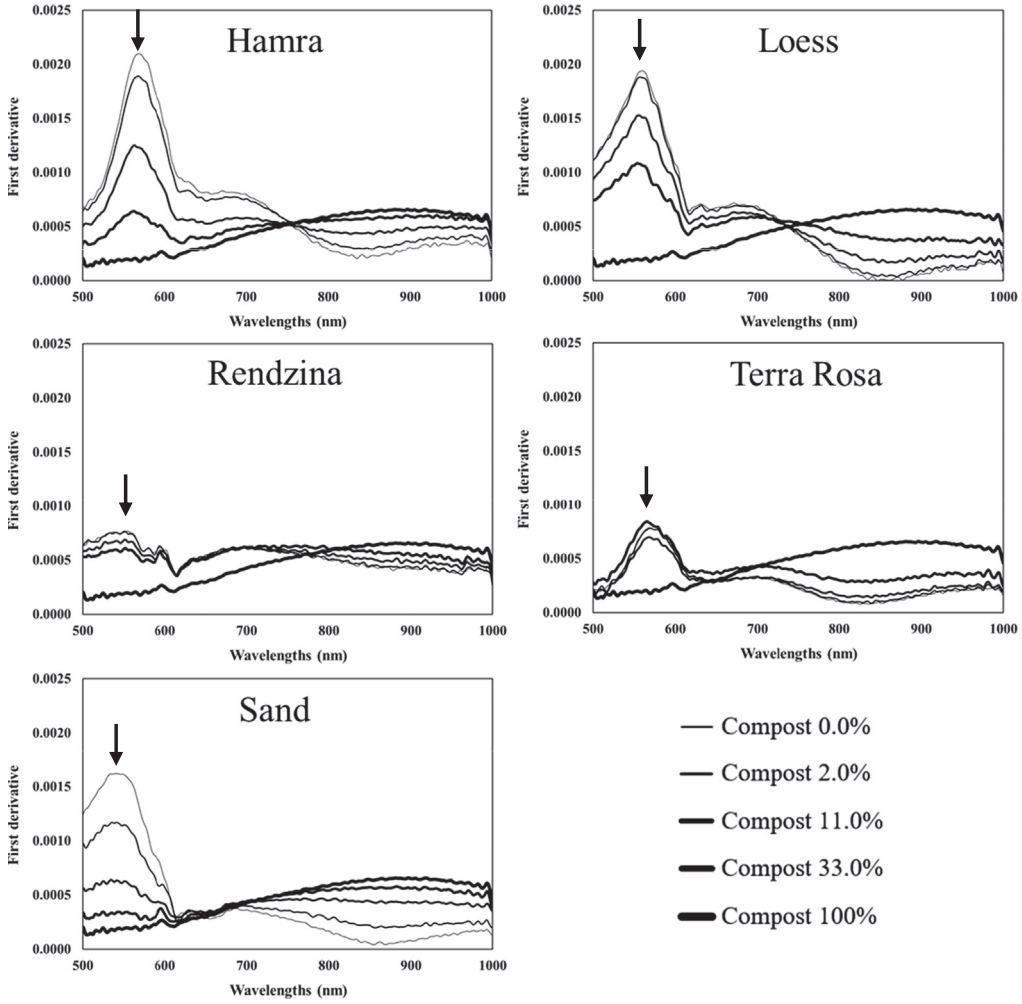


Fig. 6. The first-derivative spectra of soils between 500 and1000 nm with various amounts of compost. The arrow marks the location of the SIW.

(0.0–42.9% of compost).

In contrast to the variance in the R^2 results obtained with the ASD, for the HySpex data (Table 3b), all soils reached their highest R^2 only when using all 18 samples (0.0–42.9% of compost), except for Sand which reached its maximum with 14 samples (0.0–5.0% of compost). This difference might be due to the lower spectral sampling intervals of the HySpex and the transition from point spectroscopy to imaging data.

The semi-transparent Sand medium showed a sharp decrease in R^2 in both datasets. The Sand-OM relation appears to be linear, then shifts to a non-linear pattern after reaching a certain amount of OM. On the other hand, the R^2 of Hamra soil stayed consistent, probably because the original Sand particles are coated with Fe-oxides and thus, it is not as transparent as the Sand. Figs. 7 and 8 show the highest correlations using ASD and HySpex sensors, respectively. The number of samples required for the DL assessment varied between soil types and between the two sensors.

Results are summarized in Table 4 where the uncertainty range—the anomaly range (%) from the OM_S—is also given. Regarding the ASD sensor, the lowest $SSDL_{OM}$ was observed in the Sand, followed by Hamra. These two samples originate from coarse quartz particles,

Table 3
Summary of the linear coefficient of determination values between SIW/CIW and compost from 10 samples (minimum required) to 18 samples for ASD (a) and HySpex (b) sensors.

		Number of soil samples								
		10	11	12	13	14	15	16	17	18
a	Hamra	**0.933**	0.915	0.859	0.863	0.830	0.887	0.877	0.841	0.831
	Loess	0.762	0.781	0.797	**0.828**	0.764	0.713	0.695	0.666	0.676
	Rendzina	0.238	0.398	0.384	0.577	0.576	0.846	0.887	0.898	**0.913**
	Terra Rosa	0.341	0.336	0.431	0.288	0.544	0.833	0.815	0.883	**0.908**
	Sand	0.860	**0.874**	0.816	0.814	0.682	0.490	0.429	0.397	0.406
b	Hamra	0.840	0.861	0.803	0.785	0.862	0.845	0.864	0.863	**0.870**
	Loess	0.408	0.346	0.481	0.633	0.705	0.772	0.829	0.836	**0.847**
	Rendzina	0.126	0.161	0.378	0.323	0.364	0.831	0.909	0.911	**0.918**
	Terra Rosa	0.405	0.409	0.476	0.055	0.016	0.331	0.291	0.718	**0.850**
	Sand	0.726	0.776	0.772	0.801	**0.811**	0.678	0.633	0.601	0.614

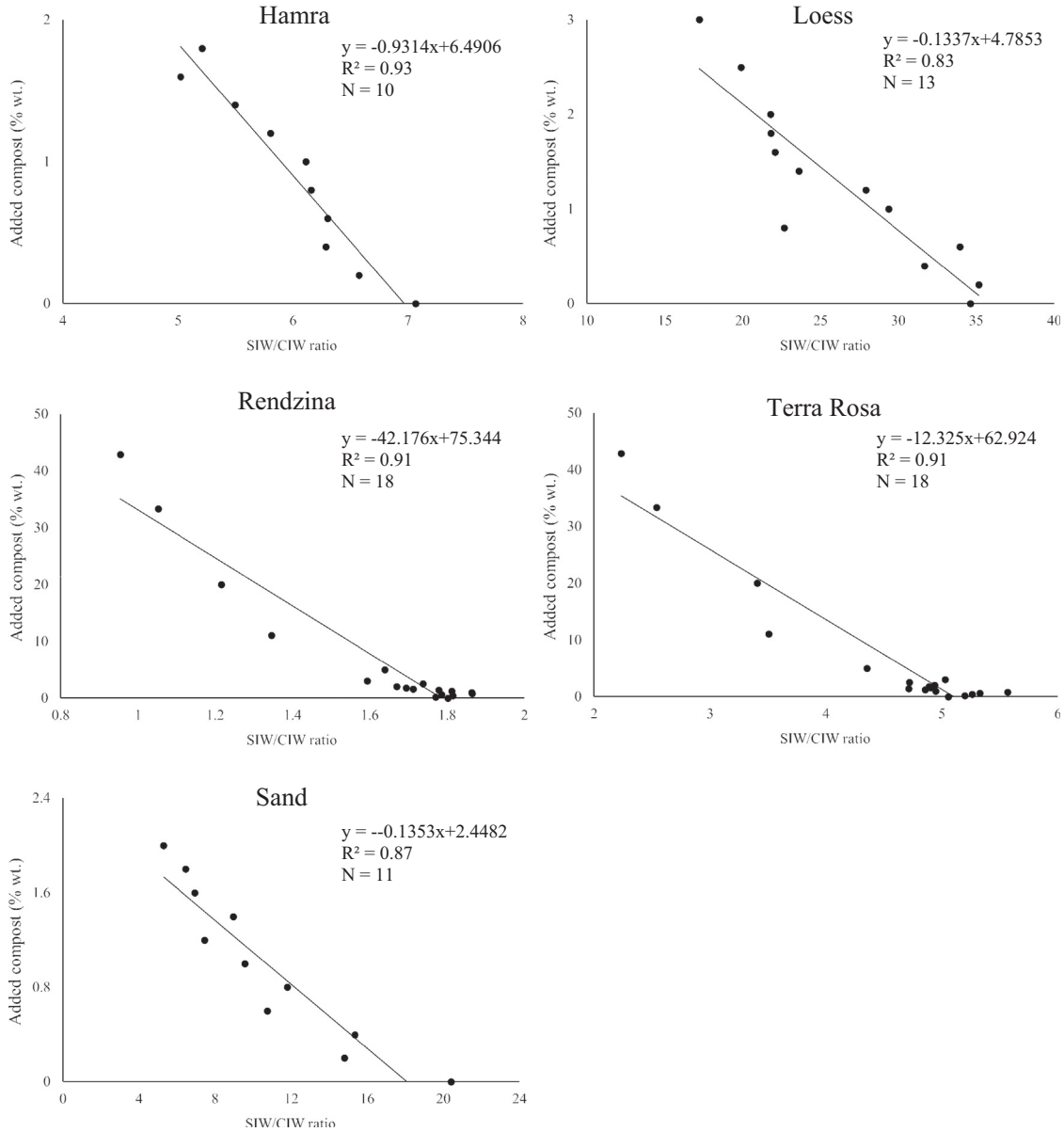


Fig. 7. Linear correlations between SIW/CIW ratio and the added compost for all soil types using the ASD spectrometer.

contain low amounts of OM_S and no clay particles. The two samples also showed the lowest $SSDL_{OM}$ anomaly which caused an uncertainty range of ± 28% and ± 21% for the Sand and the Hamra samples, respectively. The Loess sample (loamy texture) contained higher amounts of OM_S and showed a slightly higher anomaly, which translated to ± 19% uncertainty range. Rendzina and Terra Rosa samples have a silty loam to clay texture and contain higher amounts of OM_S than the other soils. The two samples showed the highest $SSDL_{OM}$ anomaly, causing an uncertainty range of ± 25% and ± 28% for Rendzina and Terra Rosa, respectively.

HySpex results exhibited a similar trend, with Sand and Hamra showing the lowest anomalies followed by Loess, Rendzina and Terra Rosa samples. The only difference was the value of their anomalies, which was much higher than in the ASD data. Sand and Hamra showed the lowest anomalies with ± 327% and ± 362% uncertainty range, respectively. In contrast, Loess, Rendzina and Terra Rosa showed higher anomalies with 114%, 123% and 118% uncertainty ranges.

The dissimilarities between the results achieved with the ASD and the HySpex can result from the measurement method (contact/image), size of the sampled area and the spectral resolution.

In addition, we examined the relationship between ASD and HySpex $SSDL_{OM}$ anomaly results. Fig. 9 shows a high correlation between the two datasets, which tells us that our spectral measurements were accurate and the method that we used applicative for both sensors, dependent upon soil type. Moreover, despite performing all measurements in laboratory conditions, the spectral performance of the ASD was almost four times (3.7737) higher than the HySpex images (Fig. 9) which may have been stemmed from the spectral resolution of the analyzed dataset (1 nm vs. 3.6 nm in the ASD and HySpex, respectively). However, knowing the $SSDL_{OM}$ of the ASD enabled estimating it in the HySpex sensor despite their spectral and spatial differences.

3.3. The contribution of clay and OM_S on $SSDL_{OM}$ assessment

A question arose as to the contribution of the clay content and OM_S to the $SSDL_{OM}$ assessment. To answer this, we applied a multivariate correlation using the SPSS statistical software (IBM Corp., 2011). Although only 5 samples were used for the statistics, we obtained preliminary and exploratory insight into the ability to predict SSDL using varied sensors.

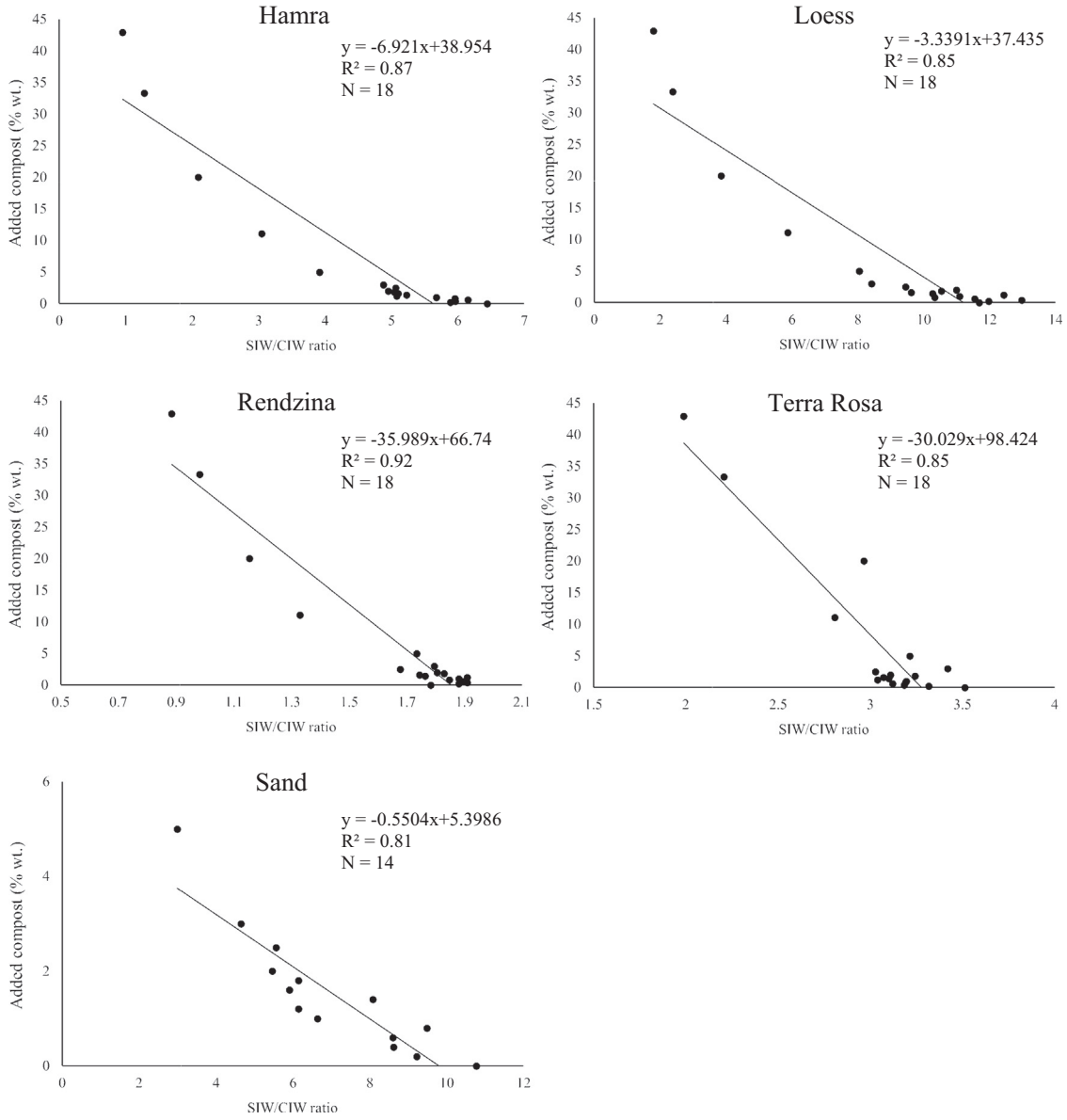


Fig. 8. Linear correlations between SIW/CIW ratio and the added compost for all soil types using the HySpex sensor.

A pairwise Pearson correlation was performed between clay and OM_S in order to identify if they are orthogonal. The correlation between the two variables was 0.96, the $R^2 = 0.93$ and the adjusted $R^2 = 0.90$. ASD results showed that high SSDL anomaly values correspond to high clay and OM_S contents with $R^2 = 0.925$ and $R^2 = 0.987$, respectively. The multivariable correlation for both clay and OM_S contents for predicting the SSDL anomaly resulted in $R^2 = 0.984$. HySpex results showed a higher correlation between SSDL anomaly and clay ($R^2 = 0.943$), and between the former and OM_S ($R^2 = 0.993$), compared to the ASD sensor. The multivariable correlation of both predictors resulted in $R^2 = 0.988$. The influence of adding a certain amount of OM and detecting it with any hyperspectral sensor, is related mostly to OM_S. The percentage of OM needed to reach the $SSDL_{OM}$ (uncertainty range) is approximately the same for all soil types using the ASD (19–28%) and varies using the HySpex (114–362%), as presented in Table 4. This means that any spectral models which predict

Table 4
Summary of the $SSDL_{OM}$ derived from ASD and HySpex sensors.

	OM_S (% wt)	ASD				HySpex			
		Lower SSDL (% wt)	Upper SSDL (% wt)	$SSDL_{OM}$ anomaly	Uncertainty range (%)	Lower SSDL (% wt)	Upper SSDL (% wt)	$SSDL_{OM}$ anomaly	Uncertainty range (%)
Sand	0.27	0.19	0.35	0.08	± 28	0	1.15	0.88	± 327
Hamra	0.61	0.48	0.74	0.13	± 21	0	2.82	2.21	± 362
Loess	4.13	3.35	4.91	0.78	± 19	0	8.87	4.74	± 114
Rendzina	5.23	3.92	6.53	1.30	± 25	0	11.68	6.45	± 123
Terra Rosa	9.01	6.49	11.53	2.52	± 28	0	19.64	10.63	± 118

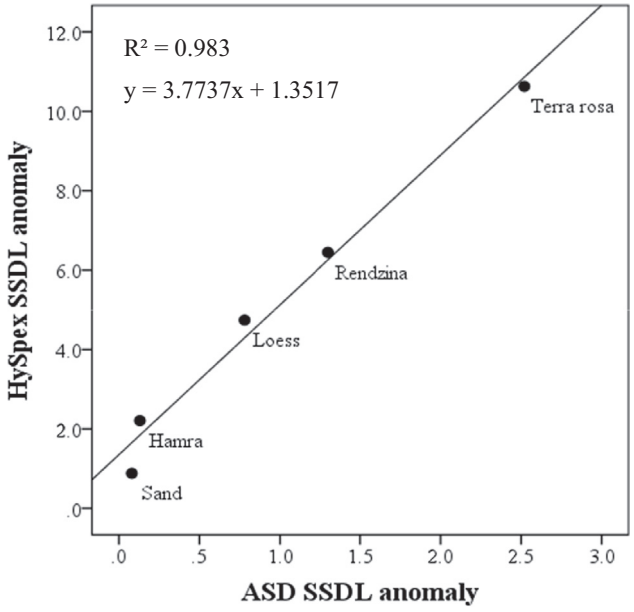


Fig. 9. Correlation between SSDL anomalies of ASD and HySpex.

OM content that falls in the uncertainty range should be neglected and should not be added to the prediction model.

3.4. Partial least squares (PLS) models

After evaluating the DL for each soil, we demonstrated the ability to build an OM_{SC} model with the commonly used PLS method with the PLS_Toolbox (2016), using 5 latent variables and the venetian blinds for cross-validation. For model evaluation, we used the root mean squares errors of calibration (RMSEC), cross-validation (RMSECV) and prediction (RMSEP) together with their coefficient of determination (R^2) values. The high accuracy and spectral resolution of the ASD measurements enabled distinguishing the soil samples, even though there were only small differences in the OM content, and each soil type showed a specific trend in the generic model. For example, in Fig. 10a, the sandy samples of Hamra and Sand presented a similar trend, which differed from those shown by Loess, Rendzina and Terra Rosa. In general, two linear trends were visible, with the 2% value marking the threshold; samples containing < 2% OM_{SC} and samples containing over 2% OM_{SC}. The steep trend of the samples containing < 2.5% OM_{SC} (Hamra and Sand) might be due to the presence of OM_{SC} particles which are visible in between the coarse quartz particles. On the other hand, samples containing > 2.5% OM_{SC}, the slope of the linear curve is more gradual. In addition, the generic model underestimated OM_{SC} content, followed by overestimation of high OM_{SC} contents in Hamra and Sand, and underestimation in Loess, Rendzina and Terra Rosa. The difference is might be due to the formation of micro-aggregates in soils with higher contents of clay and silt (Edwards and Bremner, 1967). In this process, OM and clay are linked by polyvalent metals, which may affect the appearance of OM in the spectral data.

The HySpex sensor is less precise and less sensitive to any change in OM_{SC} values, resulting in the formation of clusters rather than trends, as shown in Fig. 10b. The gradual trends that were observed in each soil type using the ASD data disappeared, the linear trends within each soil type were not observed, and the overall linearity was caused only by the presence of the clusters.

The two linear regressions supported the notion that when using data obtained from the ASD, a certain trend is visible in each soil type, but that this is disrupted when using the imaging data of the HySpex sensor. This means that a prediction model using any imaging data might perform well because it performs an estimation through an

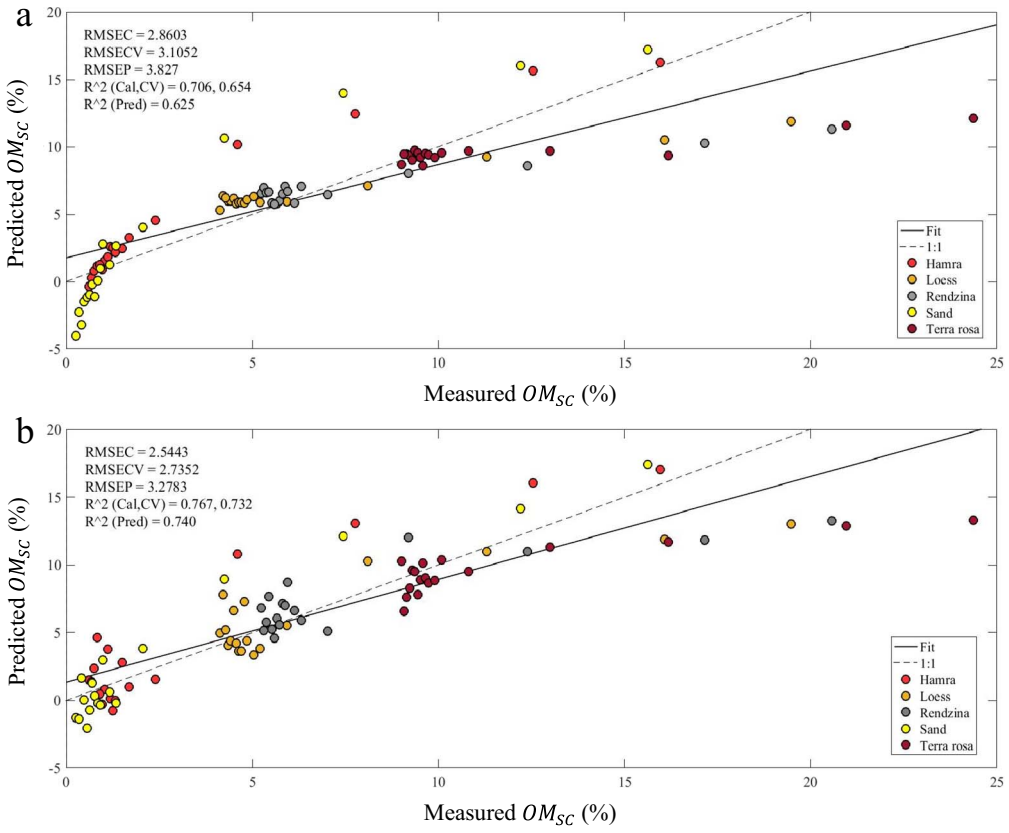


Fig. 10. Prediction models of organic matter which include both calibration and validation samples for ASD (a) and HySpex (b) sensors.

indirect assessment of spectral clustering and linearity exists only when several soil types are combined.

4. Conclusions

This work highlights the problematic and tricky assessment of SOM performed today using spectral measurements and statistical methods. In recent years, soil spectral analyses have converged into the performance of generic models (which might perform well in certain cases), but this study shows that to accurately predict SOM, it is essential to take into account the soil type and its DL values. Each soil type was found to exhibit different $SSDL_{OM}$ and variant uncertainty ranges with respect to the prediction of OM and in addition, the sensor itself has a major impact on the DL threshold. Consequently, building a database containing diverse soil types might not be the right method for spectral analysis and generic model use.

To overcome this misleading and fallacious assessment, it is advisable to avoid generic models and instead, perform a preliminary classification of the dataset, as was suggested by Ogen et al., 2017, and then build an exclusive prediction model which includes soils with similar spectral characteristics. Subsequently, for each model, samples with OM_S values that are under the DL should be identified and marked as samples with limited liability. This will ensure a proper spectral analysis, which will make prediction models more accurate and reliable. The main conclusions of this study are:

1. The SIW/CIW ratio was found to be a reliable and accurate method for $SSDL_{OM}$ assessment using ASD and HySpex sensors; SIW and CIW values were dependent on soil sensor.
2. ASD and HySpex sensors showed a correlation between $SSDL_{OM}$ anomaly results. Knowing one sensor's $SSDL_{OM}$ enables estimating that of the other sensor.
3. Soil types differed in their $SSDL_{OM}$ values, and this was correlated to the OM_S and clay contents.
4. The spectral modeling of OM depends on the soil type and the initial OM content in a certain field. Thus, it is recommended to avoid applying the models on other areas.
5. A generic prediction model using the imaging spectroscopy, provides a rough estimation of SOM content due to spectrum affiliation into a certain class and not to a correct chemometric approach. Therefore, prior classification of the dataset is advisable for a more accurate prediction of SOM content.

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Author contributions

Y.O. prepared the soil samples, processed and analyzed the spectral data, and is the main author of the article. C.N. acquired the imaging data. S.C., N.G. and E.B.D. were the project supervisors.

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